

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTrainingData/Static · 425 trials
Total training paths	1 · 425 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/StaticReports

Sensor selection

Hands	left, right
Segments	thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	164 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	none
Feature mode	stats → feature dim 1312
Test size · seed · CV folds	0.2 · random_state=42 · 5-fold stratified
Sample counts	train (post-aug)=340 · test=85

Augmentation (training only)

Enabled	no
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (5-fold)	CV std	Hold-out test acc
SVM (Linear)	0.9941	0.0072	1.0000
Random Forest	0.9971	0.0059	1.0000
Logistic Regression	0.9971	0.0059	1.0000
LDA	0.9882	0.0110	1.0000
KNN	0.9441	0.0253	0.9647
SVM (RBF)	0.9676	0.0285	0.9529

Per-class hold-out F1

Class	SVM (Linear)	Random Forest	Logistic Regression	LDA	KNN	SVM (RBF)	Support
Double_Closed_Fist	1.000	1.000	1.000	1.000	0.963	0.929	13
Double_Nothing	1.000	1.000	1.000	1.000	0.957	0.917	12
Double_Okay	1.000	1.000	1.000	1.000	1.000	1.000	12
Double_Open_Palm	1.000	1.000	1.000	1.000	1.000	1.000	12
Double_Phone	1.000	1.000	1.000	1.000	0.917	0.909	12
Double_Pistol	1.000	1.000	1.000	1.000	0.957	0.957	12
Double_Spiderman	1.000	1.000	1.000	1.000	0.960	0.960	12
macro avg	1.000	1.000	1.000	1.000	0.965	0.953	85

External test (NewTestData)

Algorithm	External test acc
Random Forest	0.5833
LDA	0.4444
SVM (RBF)	0.3611
KNN	0.3611
Logistic Regression	0.3333
SVM (Linear)	0.3056

Per-class external accuracy

Class	Random Forest	LDA	SVM (RBF)	KNN	Logistic Regression	SVM (Linear)	Support
Double_Closed_Fist	0.333	0.000	0.000	0.000	0.000	0.000	6
Double_Nothing	0.800	1.000	0.600	0.000	0.200	0.200	5
Double_Okay	0.000	0.000	0.000	0.000	0.000	0.000	5

Double_Open_Palm	0.000	0.000	0.000	0.000	0.000	0.000	5
Double_Phone	1.000	0.200	0.000	0.600	0.800	0.000	5
Double_Pistol	1.000	1.000	1.000	1.000	1.000	1.000	5
Double_Spiderman	1.000	1.000	1.000	1.000	0.400	1.000	5